



Abhishek Aylaveni

Embedded Systems / IoT / Electronic Validation Internship

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Professional Profile

Objective: embedded systems / IoT / electronic validation engineering internship. Master's student in Electronics and Embedded Systems at ESIGELEC, focused on embedded firmware, sensors, wireless communication, real-time systems, and testable prototypes. Public portfolio and GitHub document selected projects, technical stack, and validation work. Skills: embedded C, STM32, ESP32, ARM Cortex-M, LoRaWAN, MQTT, UART/SPI/I2C/CAN, RTOS / FreeRTOS, hardware debugging, and functional testing.

Experience

Electrical Trainee

Indian Power Projects Ltd Dec. 2022 – Nov. 2023

- Contributed to the design and testing of transformers and electrical control panels.
- Supported preventive maintenance activities, reducing unplanned downtime by 10%.
- Assisted real-time SCADA monitoring for electrical network stability.

Projects

LoRa-Based IoT Tracking System

STM32, LoRa Oct. 2025 – Dec. 2025

- Problem: need for long-range tracking with low power consumption.
- Approach: STM32 embedded C firmware, sensor integration, LoRa communication, and field testing.
- Result: achieved 5 km field range and reduced power consumption by 25%.

Autonomous Line-Following Robot

ARM Cortex-M, Embedded C Apr. 2025 – Jun. 2025

- Implemented real-time PID control for trajectory tracking, reaching 90% accuracy.
- Optimized sensor calibration and real-time processing, reducing latency by 20%.

Smart Parking IoT System

ESP / Wi-Fi, Ultrasonic Sensors Apr. 2022 – Jul. 2022

- Built a real-time IoT availability system with ultrasonic sensors, ESP / Wi-Fi, and a dashboard.
- Reduced estimated parking search time by 30% through live slot monitoring.

IoT Temperature Controller

ESP32, MQTT, Temperature Sensor Nov. 2021 – Feb. 2022

- Developed an ESP32 and MQTT connected controller to publish temperature data in real time.
- Integrated a web dashboard for monitoring, alerts, and energy consumption optimization.

Education

M.Sc. Electronics and Embedded Systems

ESIGELEC, Rouen, France 2025 – 2026

Real-time systems, microcontrollers, sensors, wireless communication, validation.

Bachelor Electrical and Electronics Engineering

TKR Engineering College, India 2018 – 2022

Electronics, electrical systems, embedded projects, and IoT.

Technical Skills

Languages: Embedded C, C, Python, test scripts

Embedded: STM32, ESP32, ARM Cortex-M, Arduino, RTOS / FreeRTOS, firmware, microcontrollers

Protocols: LoRaWAN, MQTT, CAN, UART, SPI, I2C, Wi-Fi

Validation: Oscilloscope, logic analyzer, hardware debugging, test plans, functional testing, prototype validation

Tools: Git, GitHub, VS Code, Linux, MATLAB, Simulink, technical documentation, datasheet reading

Languages

English – C2

French – A2, improving

Hindi – native



Professional Strengths

- Reading datasheets, electronic schematics, and technical documentation.
- Functional testing, hardware debugging, and prototype validation.
- Technical English communication and adaptation in multicultural teams.